Part A, Topic 1 - COVID-19

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May 30, 2025

Contents

1 Data Wrangling and Integration

First, load the packages needed.

```
options(repos = c(CRAN = "https://cloud.r-project.org/"))

if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse")
}
if (!requireNamespace("lubridate", quietly = TRUE)) {
  install.packages("lubridate")
}
if (!requireNamespace("impute", quietly = TRUE)) {
  install.packages("impute")
}
if (!requireNamespace("dplyr", quietly = TRUE)) {
  install.packages("dplyr")
}
if (!requireNamespace("magrittr", quietly = TRUE)) {
  install.packages("magrittr")
}
if (!requireNamespace("ggplot2", quietly = TRUE)) {
  install.packages("ggplot2")
}
if (!requireNamespace("readxl", quietly = TRUE)) {
  install.packages("readxl")
}
if (!requireNamespace("stringr", quietly = TRUE)) {
  install.packages("stringr")
}
if (!requireNamespace("forcats", quietly = TRUE)) {
  install.packages("forcats")
}
if (!requireNamespace("tidytext", quietly = TRUE)) {
  install.packages("tidytext")
}

library(tidyverse)
library(lubridate)
library(impute)
library(dplyr)
library(magrittr)
library(ggplot2)
library(readxl)
library(stringr)
library(forcats)
library(tidytext)
```

Then, read and open the three datasets, Covid-data.csv, CountryLockdowndates.csv and WorldwideVaccine-Data.csv

```
covid <- read.csv("Covid-data.csv")
lockdown <- read.csv("CountryLockdowndates.csv")
vaccine <- read.csv("WorldwideVaccineData.csv")
```

1.1 Exploration

Covid-data.csv

```
cat("Number of rows:", nrow(covid), "\n")
cat("Number of columns:", ncol(covid), "\n")
cat("Number of missing values:", sum(is.na(covid)), "\n")
knitr::kable(head(covid), caption = "First 6 rows of the dataset")
cat("Structure of the dataset:\n")
str(covid)
cat("Summary of the dataset:\n")
summary(covid)
```

```
## Number of rows: 1575
```

```
## Number of columns: 8
```

```
## Number of missing values: 13
```

Table 1: First 6 rows of the dataset

location	date	total_cases	new_cases	total_deaths	new_deaths	gdp_per_capita	population
Australia	2019-12-31	0	0	0	0	44648.71	25499881
Australia	2020-01-01	0	0	0	0	44648.71	25499881
Australia	2020-01-02	0	0	0	0	44648.71	25499881
Australia	2020-01-03	0	0	0	0	44648.71	25499881
Australia	2020-01-04	0	0	0	0	44648.71	25499881
Australia	2020-01-05	0	0	0	0	44648.71	25499881

```
## Structure of the dataset:
```

```
## 'data.frame':    1575 obs. of  8 variables:
##  $ location      : chr  "Australia" "Australia" "Australia" "Australia" ...
##  $ date          : chr  "2019-12-31" "2020-01-01" "2020-01-02" "2020-01-03" ...
##  $ total_cases   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ new_cases     : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ total_deaths  : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ new_deaths    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ gdp_per_capita: num  44649 44649 44649 44649 44649 ...
##  $ population    : int  25499881 25499881 25499881 25499881 25499881 25499881 25499881 25499881 2549
```

```
## Summary of the dataset:
```

```
##    location             date            total_cases         new_cases
##  Length:1575        Length:1575        Min.   :      0.0   Min.   :-29726
##  Class :character   Class :character   1st Qu.:     21.5   1st Qu.:     1
##  Mode  :character   Mode  :character   Median :  58226.0   Median :   205
##                                        Mean   : 180451.9   Mean   :  2971
##                                        3rd Qu.: 173133.0   3rd Qu.:  1880
##                                        Max.   :3363056.0   Max.   : 66625
##
##   total_deaths      new_deaths      gdp_per_capita    population
##  Min.   :     0   Min.   :-1918.0   Min.   :15309   Min.   :2.550e+07
##  1st Qu.:     0   1st Qu.:    0.0   1st Qu.:26677   1st Qu.:6.046e+07
##  Median :  2837   Median :    5.0   Median :38606   Median :6.789e+07
##  Mean   : 14060   Mean   :  183.8   Mean   :35140   Mean   :2.652e+08
##  3rd Qu.: 25100   3rd Qu.:  149.0   3rd Qu.:42201   3rd Qu.:2.075e+08
##  Max.   :135605   Max.   : 4928.0   Max.   :54225   Max.   :1.439e+09
##  NA's   :6        NA's   :7
```

CountryLockdowndates.csv

```r
cat("Number of rows:", nrow(lockdown), "\n")
cat("Number of columns:", ncol(lockdown), "\n")
cat("Number of missing values:", sum(is.na(lockdown)), "\n")
knitr::kable(head(lockdown), caption = "First 6 rows of the dataset")
cat("Structure of the dataset:\n")
str(lockdown)
cat("Summary of the dataset:\n")
summary(lockdown)
```

```
## Number of rows: 307
```

```
## Number of columns: 5
```

```
## Number of missing values: 0
```

Table 2: First 6 rows of the dataset

Country.Region	Province	Date	Type	Reference
Afghanistan		24/03/2020	Full	https://www.thestatesman.com/world/afghan-govt-imposes-lockdown-coronavirus-cases-increase-15-1502870945.html
Albania		08/03/2020	Full	https://en.wikipedia.org/wiki/2020_coronavirus_pandemic_in_Albania
Algeria		24/03/2020	Full	https://www.garda.com/crisis24/news-alerts/325896/algeria-government-implements-lockdown-and-curfew-in-blida-and-algiers-march-23-update-7
Andorra		16/03/2020	Full	https://en.wikipedia.org/wiki/2020_coronavirus_pandemic_in_Andorra
Angola		24/03/2020	Full	https://en.wikipedia.org/wiki/2020_coronavirus_pandemic_in_Angola

Country.Region	Province	Date	Type	Reference
Antigua and Barbuda			None	

```
## Structure of the dataset:

## 'data.frame':    307 obs. of  5 variables:
##  $ Country.Region: chr  "Afghanistan" "Albania" "Algeria" "Andorra" ...
##  $ Province      : chr  "" "" "" "" ...
##  $ Date          : chr  "24/03/2020" "08/03/2020" "24/03/2020" "16/03/2020" ...
##  $ Type          : chr  "Full" "Full" "Full" "Full" ...
##  $ Reference     : chr  "https://www.thestatesman.com/world/afghan-govt-imposes-lockdown-coronavirus-

## Summary of the dataset:

##  Country.Region       Province             Date               Type          
##  Length:307         Length:307         Length:307         Length:307        
##  Class :character   Class :character   Class :character   Class :character  
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character  
##   Reference        
##  Length:307        
##  Class :character  
##  Mode  :character  
```

WorldwideVaccineData.csv

```r
cat("Number of rows:", nrow(vaccine), "\n")
cat("Number of columns:", ncol(vaccine), "\n")
cat("Number of missing values:", sum(is.na(vaccine)), "\n")
knitr::kable(head(vaccine), caption = "First 6 rows of the dataset")
cat("Structure of the dataset:\n")
str(vaccine)
summary(vaccine)
```

```
## Number of rows: 187

## Number of columns: 5

## Number of missing values: 0
```

Table 3: First 6 rows of the dataset

Country	Doses.administered.per.100.people	Total.doses.administered	X..of.population.vaccinated	X..of.population.fully.vaccinated
Afghanistan	17	6445359	15	13
Albania	102	2906126	46	44

Country	Doses.administered.per.100.people	Total.doses.administered	X..of.population.vaccinated	X..of.population.fully.vaccinated
Algeria	35	15205854	19	16
Angola	64	20397115	41	22
Argentina	237	106474858	92	84
Armenia	73	2150112	38	33

```
## Structure of the dataset:

## 'data.frame':    187 obs. of  5 variables:
##  $ Country                          : chr  "Afghanistan" "Albania" "Algeria" "Angola" ...
##  $ Doses.administered.per.100.people: int  17 102 35 64 237 73 162 229 207 137 ...
##  $ Total.doses.administered         : num  6.45e+06 2.91e+06 1.52e+07 2.04e+07 1.06e+08 ...
##  $ X..of.population.vaccinated      : num  15 46 19 41 92 38 84 88 77 53 ...
##  $ X..of.population.fully.vaccinated: num  13 44 16 22 84 33 78 86 75 48 ...

##    Country          Doses.administered.per.100.people Total.doses.administered
##  Length:187         Min.   :  0                       Min.   :1.714e+04
##  Class :character   1st Qu.: 62                       1st Qu.:1.810e+06
##  Mode  :character   Median :130                       Median :8.179e+06
##                     Mean   :131                       Mean   :6.493e+07
##                     3rd Qu.:199                       3rd Qu.:2.865e+07
##                     Max.   :343                       Max.   :3.408e+09
##  X..of.population.vaccinated X..of.population.fully.vaccinated
##  Min.   : 0.10               Min.   : 0.10
##  1st Qu.:36.50               1st Qu.:29.00
##  Median :62.00               Median :55.00
##  Mean   :56.91               Mean   :51.94
##  3rd Qu.:80.00               3rd Qu.:75.00
##  Max.   :99.00               Max.   :99.00
```

1.2 Wrangling

I have chosen to use KNN imputation to deal with missing values or wrong values such as negative values for new cases, deaths, vaccinated, etc. From the exploration above, only covid has missing values, so I will impute that dataset using K Nearest Neighbors (KNN). By using KNN, I can fill in the missing values based on the values of the nearest neighbors in the dataset, allowing us keep the rest of the data for those entries.

```r
library(impute)
library(dplyr)
library(magrittr)

cat("Number of missing values before imputation:", sum(is.na(covid)), "\n")

covid_imputed <- covid %>%
  select(where(is.numeric)) %>%
  as.matrix() %>%
```

```
  impute.knn(k = 5) %>%
  .$data %>%
  as.data.frame()

covid[, colnames(covid_imputed)] <- covid_imputed

cat("Number of missing values after imputation:", sum(is.na(covid)), "\n")
```

```
## Number of missing values before imputation: 13
```

```
## Cluster size 1575 broken into 1378 197
## Done cluster 1378
## Done cluster 197
```

```
## Number of missing values after imputation: 0
```

I also want to remove any negative values from the dataset as they are not valid in some of the available features. First, check if there are indeed any entries that have such values.

Covid-data.csv

```
##   total_cases new_cases total_deaths new_deaths gdp_per_capita population
## 1       FALSE      TRUE        FALSE       TRUE          FALSE      FALSE
```

In this case, new_cases and new_deaths are the only columns that have negative values, but those values are not valid. I will remove any rows that have negative values in those columns.

```
cat("Number of rows before removing negative values:", nrow(covid), "\n")
covid <- covid %>%
  filter(new_cases >= 0 & new_deaths >= 0)
cat("Number of rows after removing negative values:", nrow(covid), "\n")
```

```
## Number of rows before removing negative values: 1575
```

```
## Number of rows after removing negative values: 1568
```

Since CountryLockdowndates.csv doesn't have numerical values, I will now proceed to WorldwideVaccine-Data.csv.

WorldwideVaccineData.csv

```
##   Doses.administered.per.100.people Total.doses.administered
## 1                             FALSE                    FALSE
##   X..of.population.vaccinated X..of.population.fully.vaccinated
## 1                       FALSE                             FALSE
```

With this, I can confirm that there are no negative values in this dataset, so no further action is needed.

1.3 Integration

Now that I have cleaned the datasets, we can integrate them into one. Firstly, the date attributes are in different formats for Covid-data.csv and CountryLockdowndates.csv, so I need to convert them to the same format.

```
covid$date <- as.Date(covid$date, format = "%Y-%m-%d")
lockdown$Date <- as.Date(lockdown$Date, format = "%d/%m/%Y")

cat("Covid-data.csv date format: ", class(covid$date), "\nHead: ", head(covid$date))
cat("CountryLockdowndates.csv date format: ", class(lockdown$Date), "\nHead: ", head(lockdown$Date))
```

```
## Covid-data.csv date format:  Date
## Head:  18261 18262 18263 18264 18265 18266
```

```
## CountryLockdowndates.csv date format:  Date
## Head:  18345 18329 18345 18337 18345 NA
```

Next, I will rename the columns in the datasets as per the requirements of the merged dataset.
The expected columns are:
- location
- date
- total_cases
- new_cases
- total_deaths
- new_deaths
- gdp_per_capita
- population
- lockdown_date
- total doses administered
- % of population fully vaccinated

```
lockdown <- lockdown %>%
  rename(location = Country.Region,
         lockdown_date = Date)
vaccine <- vaccine %>%
  rename(location = Country,
         `total doses administered` = `Total.doses.administered`,
         `% of population fully vaccinated` = `X..of.population.fully.vaccinated`)
```

Now I can join the three datasets together and review any further preprocessing that needs to be done. I will use left_join to keep all the rows from the covid dataset where the country is also present in the other two datasets. Based on the visualisation and analysis that will be conducted later, there is no need to keep entries in the other two datasets where the country is not present in the covid dataset.

```
data <- covid %>%
  left_join(lockdown %>% select(location, lockdown_date), by = "location") %>%
  left_join(vaccine %>% select(location, `total doses administered`, `% of population fully vaccinated`
```

```
cat("Number of rows:", nrow(data), "\n")
cat("Number of columns:", ncol(data), "\n")
cat("Number of missing values:", sum(is.na(covid %>% select(where(is.numeric)))), "\n")
knitr::kable(head(data), caption = "First 6 rows of the merged dataset")
cat("Structure of the merged dataset:\n")
str(data)
cat("Summary of the merged dataset:\n")
summary(data)
```

```
## Number of rows: 12303

## Number of columns: 11

## Number of missing values: 0
```

Table 4: First 6 rows of the merged dataset

location	date	total_cases	new_cases	total_deaths	new_deaths	gdp_per_capita	population	lockdown_date	total doses administered	% of population fully vaccinated
Australia	2019-12-31	0	0	0	0	44648.71	25499881	NA	57988175	86
Australia	2019-12-31	0	0	0	0	44648.71	25499881	NA	57988175	86
Australia	2019-12-31	0	0	0	0	44648.71	25499881	NA	57988175	86
Australia	2019-12-31	0	0	0	0	44648.71	25499881	NA	57988175	86
Australia	2019-12-31	0	0	0	0	44648.71	25499881	NA	57988175	86
Australia	2019-12-31	0	0	0	0	44648.71	25499881	NA	57988175	86

```
## Structure of the merged dataset:

## 'data.frame':    12303 obs. of  11 variables:
##  $ location                       : chr  "Australia" "Australia" "Australia" "Australia" ...
##  $ date                           : Date, format: "2019-12-31" "2019-12-31" ...
##  $ total_cases                    : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ new_cases                      : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ total_deaths                   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ new_deaths                     : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ gdp_per_capita                 : num  44649 44649 44649 44649 44649 ...
##  $ population                     : num  25499881 25499881 25499881 25499881 25499881 ...
##  $ lockdown_date                  : Date, format: NA NA ...
##  $ total doses administered       : num  5.8e+07 5.8e+07 5.8e+07 5.8e+07 5.8e+07 ...
##  $ % of population fully vaccinated: num  86 86 86 86 86 86 86 86 86 86 ...

## Summary of the merged dataset:

##    location              date             total_cases        new_cases
##  Length:12303       Min.   :2019-12-31   Min.   :      0   Min.   :    0.0
##  Class :character   1st Qu.:2020-02-17   1st Qu.:    882   1st Qu.:    4.0
##  Mode  :character   Median :2020-04-06   Median :  80932   Median :   43.0
##                     Mean   :2020-04-05   Mean   :  80965   Mean   :  920.1
##                     3rd Qu.:2020-05-25   3rd Qu.:  84378   3rd Qu.:  545.0
##                     Max.   :2020-07-14   Max.   :3363056   Max.   :66625.0
##                     NA's   :1189
##   total_deaths      new_deaths      gdp_per_capita    population
##  Min.   :     0   Min.   :   0.00   Min.   :15309   Min.   :2.550e+07
##  1st Qu.:    17   1st Qu.:   0.00   1st Qu.:15309   1st Qu.:6.527e+07
##  Median :  3241   Median :   1.00   Median :15309   Median :1.439e+09
##  Mean   :  7304   Mean   :  85.11   Mean   :27512   Mean   :7.624e+08
##  3rd Qu.:  4639   3rd Qu.:  37.00   3rd Qu.:39753   3rd Qu.:1.439e+09
##  Max.   :135605   Max.   :4928.00   Max.   :54225   Max.   :1.439e+09
##
##  lockdown_date        total doses administered % of population fully vaccinated
##  Min.   :2020-01-23   Min.   : 57988175        Min.   :67.00
##  1st Qu.:2020-01-23   1st Qu.: 57988175        1st Qu.:79.00
##  Median :2020-01-23   Median :146731437        Median :79.00
##  Mean   :2020-02-08   Mean   :131644832        Mean   :80.97
##  3rd Qu.:2020-03-14   3rd Qu.:146731437        3rd Qu.:86.00
##  Max.   :2020-03-27   Max.   :597655035        Max.   :86.00
##  NA's   :3840         NA's   :8078             NA's   :8078
```

```r
unique(data$location)
```

```r
unique(data$location)
```

```
##  [1] "Australia"      "Australia  "    "China"           " China"
##  [5] "France"         "Iran"           "iran"            "Italy"
##  [9] "Itly"           "Spain"          "United Kingdom" "UnitedKingdom"
## [13] "United States"  "United Stats"
```

I can see that some of the values in the location column are not consistent, so I will need to clean them up accordingly.

```
data$location <- data$location %>%
  str_replace_all("iran", "Iran") %>%
  str_replace_all("Itly", "Italy") %>%
  str_replace_all("UnitedKingdom", "United Kingdom") %>%
  str_replace_all("United Stats", "United States") %>%
  str_trim()
```

After cleaning up:

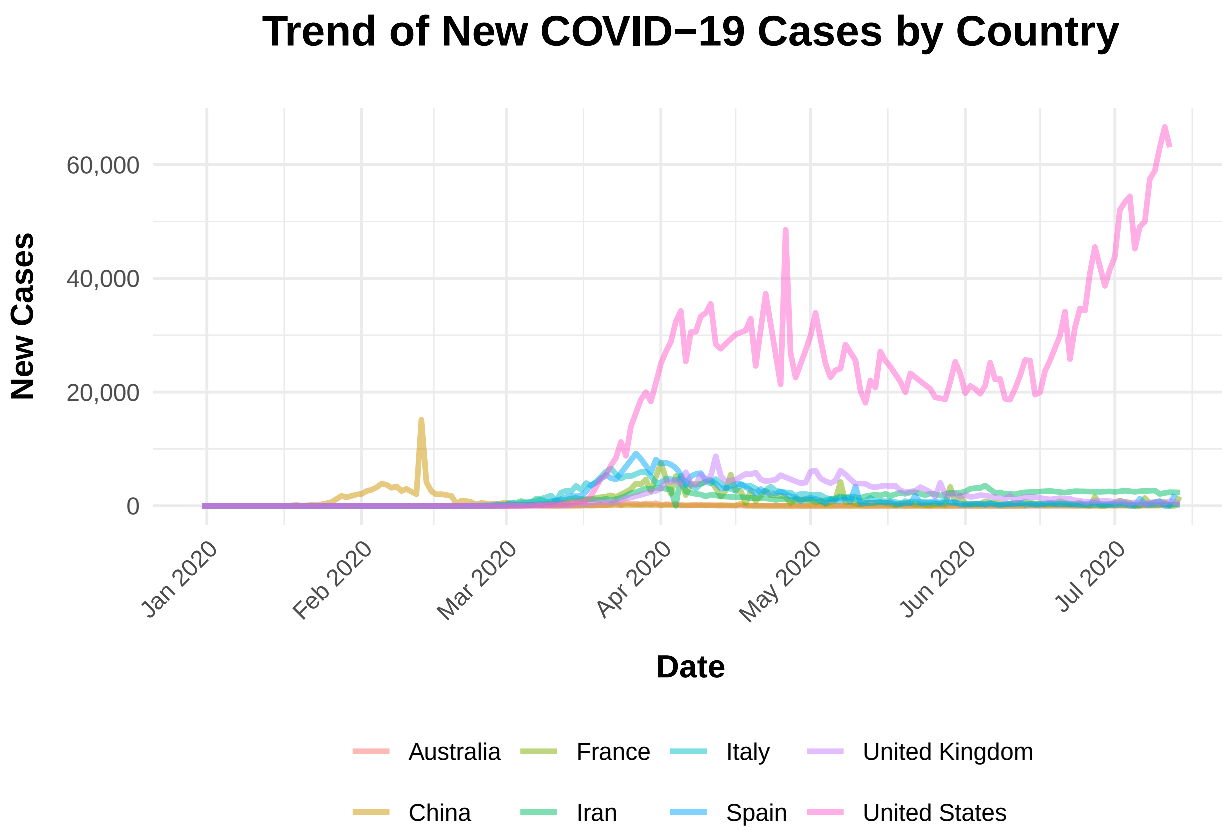
```
unique(data$location)
```

```
unique(data$location)
```

```
## [1] "Australia"      "China"          "France"         "Iran"
## [5] "Italy"          "Spain"          "United Kingdom" "United States"
```

2 Data Visualisation and Analysis

```
library(scales)

data %>%
  filter(!is.na(new_cases), !is.na(date), !is.na(location)) %>%
  ggplot(aes(x = date, y = new_cases, color = location)) +
  geom_line(size = 1, alpha = 0.5) +
  labs(title = "Trend of New COVID-19 Cases by Country", x = "Date", y = "New Cases") +
  scale_x_date(date_labels = "%b %Y", date_breaks = "1 month") +
  theme_minimal() +
  theme(
    legend.position = "bottom", legend.title = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title.x = element_text(size = 12, face = "bold", margin = margin(t = 10)),
    axis.title.y = element_text(size = 12, face = "bold", margin = margin(r = 10)),
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5, margin = margin(b = 20)),
  ) +
  scale_y_continuous(labels = comma)
```



The first occurrence of a notable observation of new cases is in China, which reflects the fact that it is the origin of the novel coronavirus outbreak, also known as COVID-19 (Centers for Disease Control and Prevention, 2024). A significant increase can also be seen in the United States, compared to other countries in the dataset, which started after about a month since China's spike. This event forced a declaration in the United States for a nationwide emergency and further overseas travel bans on the 13th of March, 2020. (Centers for Disease Control and Prevention, 2024). By August 2020, the reported total number of cases in U.S.A. was 5.04 million, which is evident of the prolonged large number of new cases daily as seen in the plot (Bergquist, S., Otten, T., & Sarich, N., 2020).

Apart from the two significant findings previously mentioned, it can also be seen that the trend of the number of new cases are similar among the European countries. By February 2020, the virus has reached Europe, with Italy being the first to react with lockdowns in different provinces as cases began to rise. By April 2020, numerous countries in Europe have reached 10,000 total cases, which is also evident in the plot (Chadwick, L., 2020)

The observed shift in the increase of new cases from China to other countries supports the fact that the outbreak originated in China. As international travel occurs daily, the spread of the virus to other places was inevitable, which explains the gradual and fluctuating rise in cases. Because of this, numerous countries implemented lockdowns and travel bans to control the spread of the virus, leading to a slow decline in new cases, which became more apparent by the middle to late part of 2020 in most countries.

Now here is an observation of the relationship between the death rate and new case numbers with the GDP of each country.

```r
mean_gdp <- data %>%
  summarise(mean_gdp = mean(gdp_per_capita, na.rm = TRUE)) %>%
  pull(mean_gdp)
cat("Mean GDP: ", mean_gdp, "\n")
```

```
## Mean GDP:  27512.46
```

```r
data <- data %>%
  mutate(GDP_Status = ifelse(gdp_per_capita >= mean_gdp, 1, 0))

cat("Number of rows with GDP Status 1: ", nrow(data[data$GDP_Status == 1, ]), "\n")
cat("Number of rows with GDP Status 0: ", nrow(data[data$GDP_Status == 0, ]), "\n")
```

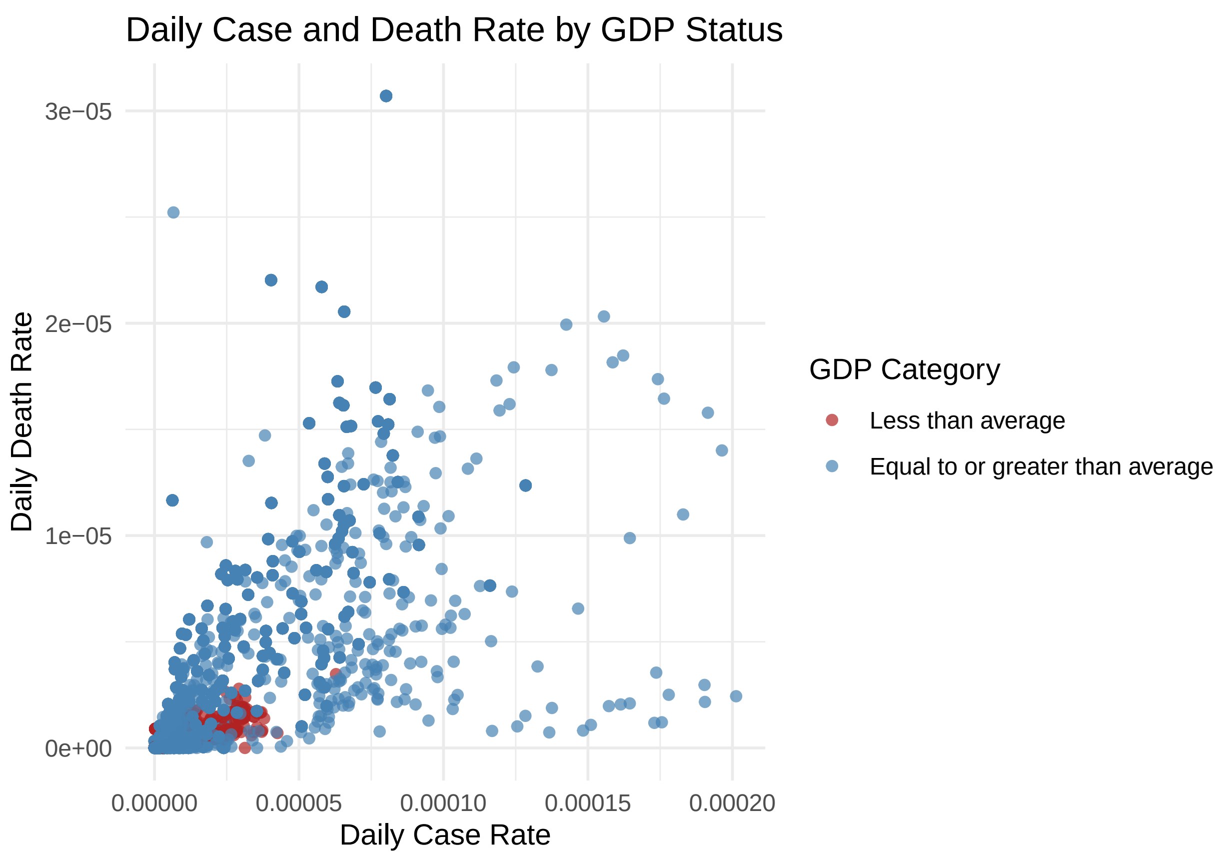
```
## Number of rows with GDP Status 1 (GDP greater than or equal to the mean):  5861

## Number of rows with GDP Status 0 (GDP less than the mean):  6442
```

```r
data <- data %>%
  mutate(new_case_rate = new_cases / population,
         new_death_rate = new_deaths / population)

head(data %>% filter(new_case_rate > 0 & new_death_rate > 0), 10)
```

```
##     location       date total_cases new_cases total_deaths new_deaths
## 1  Australia 2020-03-01          26         1            1          1
## 2  Australia 2020-03-01          26         1            1          1
## 3  Australia 2020-03-01          26         1            1          1
## 4  Australia 2020-03-01          26         1            1          1
## 5  Australia 2020-03-01          26         1            1          1
## 6  Australia 2020-03-01          26         1            1          1
## 7  Australia 2020-03-01          26         1            1          1
## 8  Australia 2020-03-01          26         1            1          1
## 9  Australia 2020-03-05          52        11            2          1
## 10 Australia 2020-03-05          52        11            2          1
##    gdp_per_capita population lockdown_date total doses administered
## 1        44648.71   25499881          <NA>                 57988175
## 2        44648.71   25499881          <NA>                 57988175
## 3        44648.71   25499881          <NA>                 57988175
## 4        44648.71   25499881          <NA>                 57988175
## 5        44648.71   25499881          <NA>                 57988175
## 6        44648.71   25499881          <NA>                 57988175
## 7        44648.71   25499881    2020-03-24                 57988175
## 8        44648.71   25499881          <NA>                 57988175
## 9        44648.71   25499881          <NA>                 57988175
## 10       44648.71   25499881          <NA>                 57988175
##    % of population fully vaccinated GDP_Status new_case_rate new_death_rate
## 1                                86          1  3.921587e-08   3.921587e-08
## 2                                86          1  3.921587e-08   3.921587e-08
## 3                                86          1  3.921587e-08   3.921587e-08
## 4                                86          1  3.921587e-08   3.921587e-08
## 5                                86          1  3.921587e-08   3.921587e-08
```

```
## 6                                    86          1  3.921587e-08   3.921587e-08
## 7                                    86          1  3.921587e-08   3.921587e-08
## 8                                    86          1  3.921587e-08   3.921587e-08
## 9                                    86          1  4.313746e-07   3.921587e-08
## 10                                   86          1  4.313746e-07   3.921587e-08
```

```
ggplot(data, aes(x = new_case_rate, y = new_death_rate, color = factor(GDP_Status))) +
  geom_point(alpha = 0.7) +
  scale_color_manual(
    values = c("0" = "firebrick", "1" = "steelblue"),
    labels = c("0" = "Less than average", "1" = "Equal to or greater than average"),
    name = "GDP Category"
  ) +
  labs(
    title = "Daily Case and Death Rate by GDP Status",
    x = "Daily Case Rate",
    y = "Daily Death Rate"
  ) +
  theme_minimal()
```



The Gross Domestic Product, or GDP, is a representation of the monetary value of goods and services produced by a country in a specified period. (Callen, T., 2025) If this value is to be divided by the total population of a country, we would get the GDP per capita, which is the number that we have in our dataset that represents a rough measure of a country's living standards, or simply, the financial capability of a person in the country.

From the scatter plot above, it can be seen that countries with lower than average GDP per capita tend to have lower daily new/death case rates. This can be due to multiple factors such as the country lacking the capability to have robust healthcare and data systems, leading to underreporting of the number of new cases and deaths per day. It can also be linked to factors such as the life expectancy and tourist spending being significantly lower than those that have average or greater GDP per capita. To provide an example:

- Luxembourg is identified as having the highest GDP per capita. Burundi is identified as having the lowest estimated GDP per capita. (The World Factbook, 2023).
- Life expectancy in Luxembourg is 82.8 years. Burundi's is 64 years. Almost 20 years difference (World Health Organisation, 2023).
- Luxembourg tourist spending for 2017 was 5.47 billion US Dollars (macrotrends, 2025). Burundi tourist spending for 2017 was 0.1 billion US dollars. (Knoema Data Appliance, 2025).

Overseas travel played a major role in the global spread of COVID-19 as rapid transmission across borders occurred. In 2020, the death rate from the virus for those that are aged 85 and over was 1,645 per 100,000 population – seven times greater than the rate for those aged 65 to 74. (Tejada-Vera, B. & Kramarow, A., 2022).

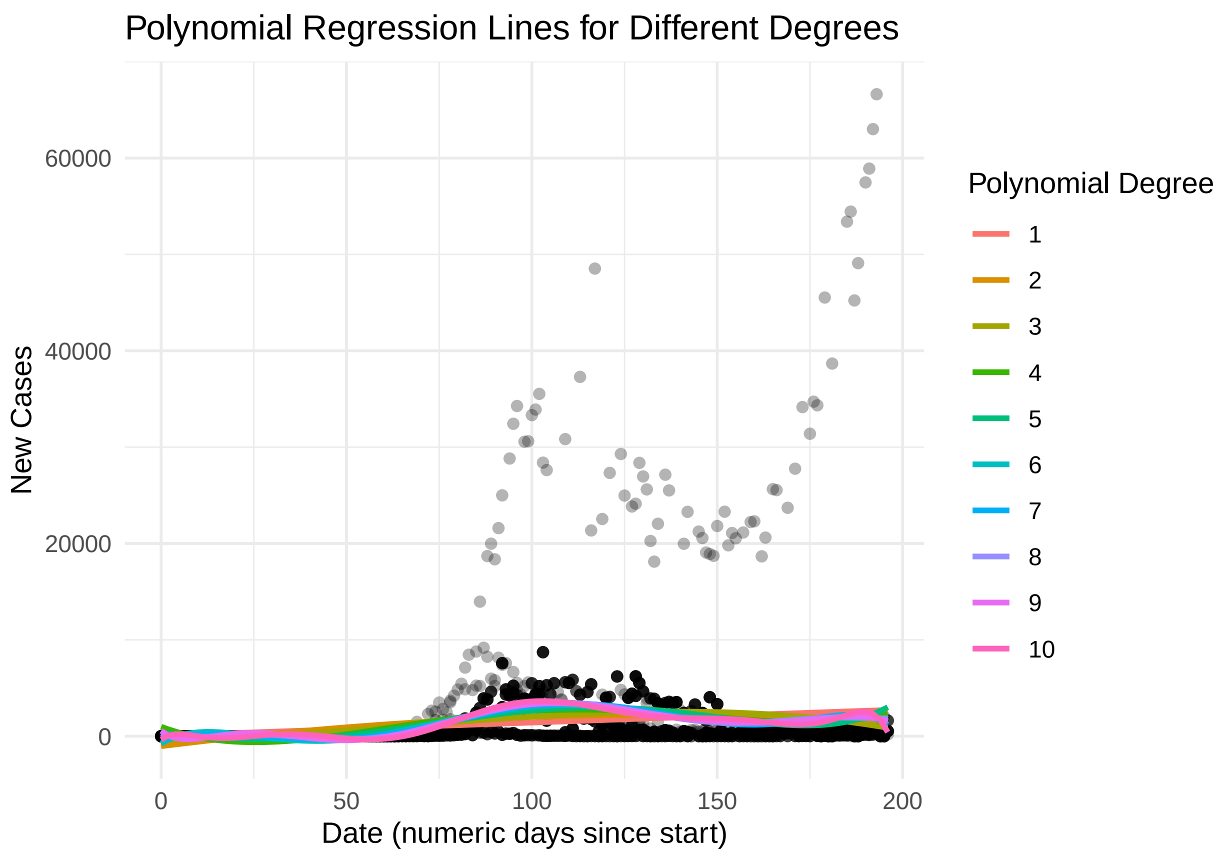
3 Predictive Data Analaysis

For this part, I will be fitting a model to make predictions on the newly infected case and death case numbers for the next five to seven days. I will be using Polynomial Regression to fit the model for the GDP group of countries whose GDP per capita is greater than or equal to the average GDP per capita of all countries in the dataset. This is because the countries in the group have daily new cases and deaths that are more spread out, which is more suitable for training a model since the data is more varied.

Firstly, let's create the model so we can plot a polynomial regression line.

```r
data_higher_gdp <- data %>%
  filter(GDP_Status == 1, !is.na(new_cases), !is.na(date)) %>%
  mutate(date_num = as.numeric(as.Date(date) - min(as.Date(date))))

# Split train/test sets
set.seed(4)
train_indices <- sample(1:nrow(data_higher_gdp), size = 0.7 * nrow(data_higher_gdp))
train <- data_higher_gdp[train_indices, ]
test <- data_higher_gdp[-train_indices, ]

# Create a sequence of dates for smooth prediction lines
date_seq <- seq(min(train$date_num), max(train$date_num), length.out = 100)
gdp_mean <- mean(train$gdp_per_capita, na.rm = TRUE)

# Initialize a dataframe to store predictions for all models
all_preds <- data.frame()

for (deg in 1:10) {
  # Fit the model with current degree for date
  model <- lm(new_cases ~ poly(date_num, deg), data = train)

  # Predict on the date sequence with fixed GDP per capita
  pred_df <- data.frame(date_num = date_seq, gdp_per_capita = gdp_mean)
```

```
  pred_df$new_cases <- predict(model, newdata = pred_df)
  pred_df$degree <- factor(deg)

  all_preds <- bind_rows(all_preds, pred_df)
}

# Plot all polynomial lines
ggplot() +
  geom_point(data = train, aes(x = date_num, y = new_cases), alpha = 0.3) +
  geom_line(data = all_preds, aes(x = date_num, y = new_cases, color = degree), size = 1) +
  labs(
    title = "Polynomial Regression Lines for Different Degrees",
    x = "Date (numeric days since start)",
    y = "New Cases",
    color = "Polynomial Degree"
  ) +
  theme_minimal()
```




```
# Model to predict new cases based on total cases
all_preds <- data.frame()  # initialize outside the loop

for (deg in 1:10) {
  model <- lm(new_cases ~ poly(total_cases, deg), data = train)

  pred_df <- data.frame(total_cases = seq(min(train$total_cases), max(train$total_cases), length.out = 
```

```
  pred_df$new_cases <- predict(model, newdata = pred_df)
  pred_df$degree <- factor(deg)

  all_preds <- bind_rows(all_preds, pred_df)
}

ggplot() +
  geom_point(data = train, aes(x = total_cases, y = new_cases), alpha = 0.3) +
  geom_line(data = all_preds, aes(x = total_cases, y = new_cases), size = 1) +
  facet_wrap(~ degree, scales = "free_x") +  # create a separate plot per degree
  labs(
    title = "Polynomial Regression Lines for Different Degrees (Total Cases)",
    x = "Total Cases",
    y = "New Cases"
  ) +
  theme_minimal()
```